higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL WORKMANSHIP
NQF LEVEL 4

20 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 Bonding is a term used to indicate the equalisation of potential✓ between two conductive bodies✓ by connecting them together with a conductor.✓ (3)

1.2 Zero potential difference (0V) (1)

1.3
- It can cause fires
- It can cause burns (2)

1.4 Water is a conductor✓ and it could be dangerous to anyone who comes in contact.✓ (2)

1.5
1.5.1 C
1.5.2 D
1.5.3 A
(3 × 1) (3)

1.6
- Verbal abuse
- Physical assault leading to bodily injuries (2)

1.7
- The safety procedure to be implemented before work is started
- The equipment to be worked on and precise location thereof
- The nature of the work to be carried out
- Identification and control of any risks arising from such work
- The person to whom the permit is issued
- Hazardous areas and points of isolation are clearly defined
- Where conductors are earthed (7)

[20]

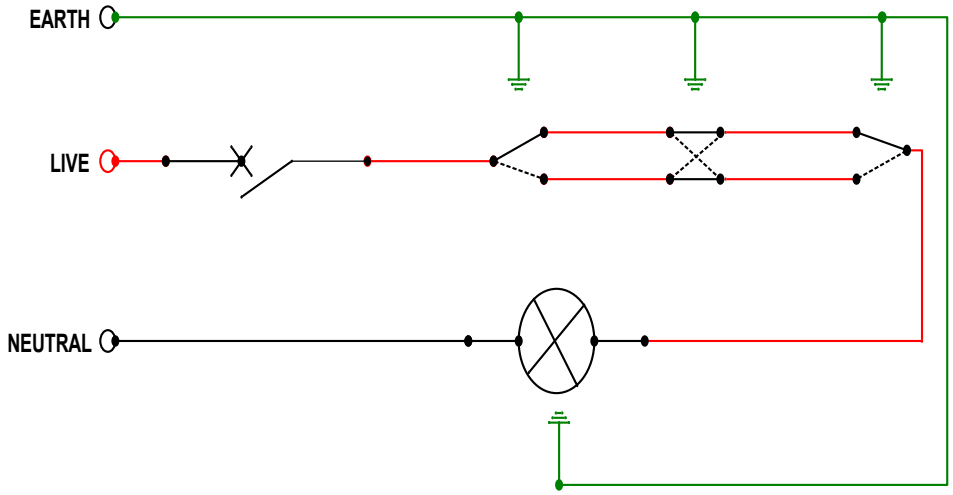
QUESTION 2

2.1 2.1.1
a installation
b connected
c protective
(3 × 1) (3)

2.1.2 Electrical separation means: the circuit is electrically isolated✓ from other circuits in an installation.✓ (2)

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2.2

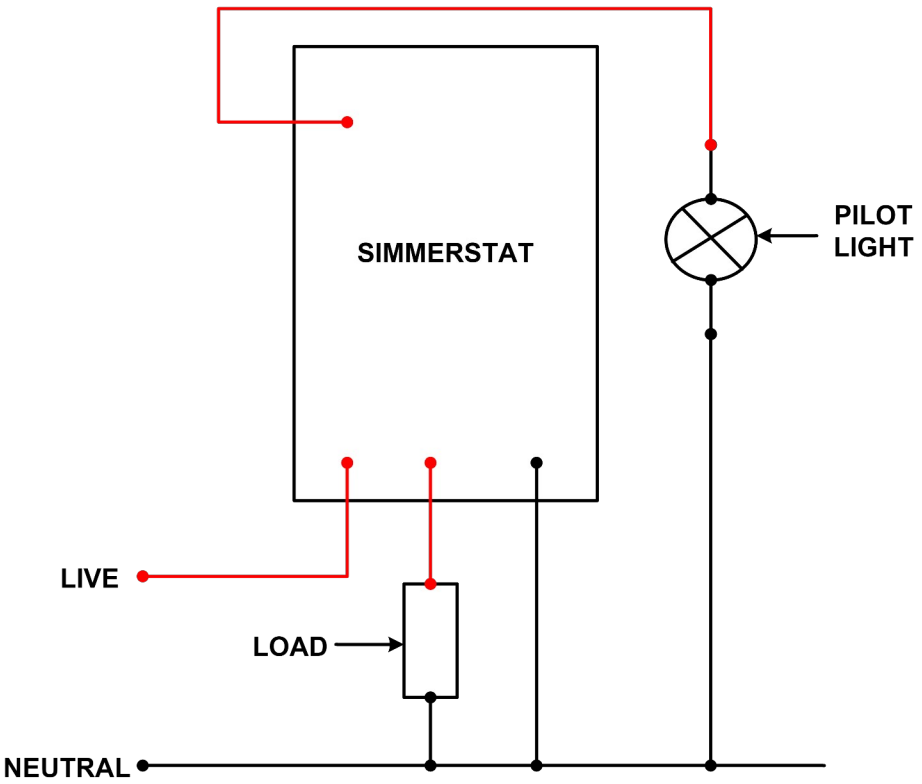


FIVE components correctly connected (5 ×1)
Earthing correctly indicated (1×1)
Any TWO correct labels of components or supply (2×1) (8)

2.3 2.3.1 Simmerstat (1)

2.3.2 It is used to control heating elements, e.g. the plates of a stove. (1)

2.3.3



Layout connections for an energy regulating switch (Simmerstat)

FOUR connections on simmerstat switch correctly indicated (4 × 1)
Any ONE correct label of connections (1 ×1) (5)

[20]

QUESTION 3

3.1 3.1.1 This is the ratio of luminous flux ✓ to electrical input power✓ (2)

3.1.2 k (1)

3.1.3 lm/W
lumens per watt (Any 1 × 1) (1)

3.2
- By using a large number of lamps
- By positioning the lamps correctly (2)

3.3 When an electric current is passed through the tungsten filament a voltage drop occurs, which consumes power.✓ This produces heat, which in turn raises the temperature of the filament wire,✓ which then causes it to glow (incandescence)✓. (3)

3.4
- They are more efficient than normal tungsten lamps.
- They last longer than normal tungsten lamps/have a long life.
- There is less depreciation in the luminous flux output during a lamp's life.
- They have a smaller light emitting area. (4)

3.5. 3.5.1 Compact fluorescent lamp (1)

3.5.2
- A gas filled tube, also called the bulb
- The magnetic or electronic ballast (2)

3.5.3
- Edison screw
- Bayonet type (2)

3.5.4
- Contain mercury
- Lamps require ballast
- Less brightness
- Initial cost is high
- Overall maintenance cost is high (Any 2 × 1) (2)

[20]

QUESTION 4

4.1
- Observation
- Asking the customer questions
- Previous fault occurrence
- Using manufacturer service manual
- Sound
- Smell
- Identification of loss of function
- Comparison
- Simulation
- Measurement (Any 5 × 1) (5)

4.2
- Test for open circuit in the start winding
- Test for open circuit in the run winding
- Test for open circuit in the centrifugal switch
- Check for dirty centrifugal switch contacts
- Check the condition of the capacitor (Any 4 × 1) (4)

4.3
- Flat twin-sheathed flex
- Two-core circular sheathed flex
- Three-core circular sheathed flex
- Unsinkable braided flex (4)

4.4
- The date of the last test
- A report of the test completed
- By whom the test was performed (3)

4.5
- Remove the tool from service.
- Replace the damaged tool immediately.
- Have the tool repaired by a qualified person.
- Follow the prescribed procedure in the workshop. (4)

[20]

QUESTION 5

5.1
- Screen protected:✓ Used in dry, dust-free conditions✓
- Flame-proof✓: Used in the mining and oil industries ✓
- Pipe-ventilated:✓ Used in a very dusty environment such as a carpentry workshop✓
- Drip-proof:✓ Used in damp conditions✓ (4 × 2) (8)

5.2
- Direct coupling
- Gear coupling
- Belt and pulley coupling
- Chain coupling
- Rigid coupling (5)

5.3
- It completely encloses the drive.
- Grills or vents for good ventilation.
- An accessible inspection door or panel.
- The guard can easily be removed and replaced if damaged.
- If necessary, the guard should also protect the drive from weather, debris and damage. (Any 4 × 1) (4)

5.4
- The stator
- The rotor
- The end shields supporting the bearings on the frame (3)

[20]

TOTAL: 100

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